Qixian ZHAO

zhao_qixian@tsinghua.edu.cn http://zhao-qixian.github.io/

Employments

2026 – present **Assistant Professor**, Yau Mathematical Sciences Center, Tsinghua University.

2023 – 2026 **Postdoctoral Fellow**, Yau Mathematical Sciences Center, Tsinghua University.
Advisor: Peng Shan.

Education

2018 – 2023 **Ph.D. Mathematics**, University of Utah.
Advisor: Dragan Miličić.

2016 – 2018 **M.A. Mathematics**, University of Wisconsin-Madison.

Spring 2015 *Concurrent Enrollment*, University of California, Berkeley.

2013 – 2017 **B.S. Mathematics and Applied Mathematics**, Sun Yat-sen University.

Publications and Preprints

[1] Taiwang Deng, Chang Huang, Bin Xu, and **Qixian Zhao**, *Relating arthur packets of real unitary groups and p-adic symplectic and orthogonal groups*, 2026. arXiv: 2603.16314 [math.RT].

[2] Peng Shan, Wenbin Yan, and **Qixian Zhao**, "Associated varieties of simple affine vertex algebras at rational levels," *Science China Mathematics*, Sep. 2026, ISSN: 1869-1862. DOI: 10.1007/s11425-026-2667-7

[3] Taiwang Deng, Chang Huang, Bin Xu, and **Qixian Zhao**, *On a geometric comparison of representations of complex and p-adic* $\mathbf{GL}_n$, 2025. arXiv: 2504.15653 [math.RT].

[4] Peng Shan, Wenbin Yan, and **Qixian Zhao**, *Cyclotomic level maps and associated varieties of simple affine vertex algebras*, 2025. arXiv: 2507.09254 [math.RT].

[5] **Qixian Zhao**, "Kazhdan-Lusztig Algorithm for Whittaker Modules With Arbitrary Infinitesimal Characters," Thesis (Ph.D.)–The University of Utah, Ph.D. dissertation, 2023, p. 106, ISBN: 979-8380-12644-1.

[6] **Qixian Zhao**, "Kazhdan-lusztig algorithm for whittaker modules with arbitrary infinitesimal characters," *Algebras and Representation Theory*, 2023, ISSN: 1572-9079. DOI: 10.1007/s10468-023-10222-0

Selected Talks

Jan 2026 **International Congress of Chinese Mathematicians (ICCM) 2025**, Shanghai, China.

Jun 2025 **Representation Theory XIX**, vertex algebra section and real group section, Inter-University Centre, Dubrovnik.

The Ninth International Conference on Representation Theory (ICRT9), Nanjing University.

May 2025 **BIMSA-YMSC Tsinghua Number Theory Seminar**, Tsinghua University.

Dec 2024 **Representation theory Seminar**, Academy of Mathematics and Systems Science, Chinese Academy of Science.

Selected Talks (continued)

Nov 2023 **Geometric Representation Theory Seminar**,
Yau Mathematical Sciences Center, Tsinghua University.

Jul 2023 **Representation theory Seminar**,
Soochow University.

Jun 2023 **Representation Theory XVIII**, real group section,
Inter-University Centre, Dubrovnik.

Apr, Oct 2019 **Graduate Student Colloquium**,
University of Utah.

Teaching

Tsinghua University

Spring 2025 Algebra 2 Honors, *TA*.

Fall 2024 Algebra 1 Honors, *TA*.

University of Utah

Spring 2023 Math 1210: Calculus I, *Instructor*.

Summer 2021 Math 2270: Linear Algebra, *Instructor*.

Fall 2019 – Fall 2022 Math 1050: College Algebra, *Instructor*.

Spring 2019 Math 1320: Engineering Calculus II, *TA*.

Fall 2018 Math 2250: Differential Equations and Linear Algebra, *TA*.

Miscellaneous

Awards

2024 **Outstanding Qiuzhen Scholar Award**,
Tsinghua University.

2023 **Outstanding Thesis Award**,
University of Utah.

2022 **Departmental Summer Research Fellowship**,
University of Utah.

2017 **Exchange & Visiting International Student Academic Excellence Award**,
University of Wisconsin-Madison.

Outstanding Thesis,
Sun Yat-sen University.

Service

2024 – 2025 **Referee** for Selecta Mathematica.

2023 – 2024 **Qiuzhen Scholar Program**, *Mentor*,
Tsinghua University.

2020 – 2021 **Graduate Student Advisory Committee Mentoring Program**, *Commitee*,
University of Utah.

2019 – 2021 **Directed Reading Program in Mathematics**, *Co-organizer*,
University of Utah.

Aug 2014 **Rural Education Support Program** at Heyuan, Guangdong, China, *Volunteered Lecturer*,
Youth League of Department of Mathematics, Sun Yat-sen University.